Question 1:
How many stars will the following code print to the monitor?

```
data = [[x for x in range(3)] for y in range(3)]
for i in range(3):
    for j in range(3):
        if data[i][j] % 2 != 0:
            print('*')
```

A. zero
B. nine
C. three
D. six

Question 2:
What is the expected output of the following code?

```
list1 = ['Peter', 'Paul', 'Mary', 'Jane']
list2 = ['Peter', 'Paul', 'Mary', 'Jane']

print(list1 is list2)
print(list1 == list2)

list1 = list2

print(list1 is list2)
print(list1 == list2)
```

A.	1 | False 2 | True 3 | False 4 | True
B.	1 | False 2 | False 3 | True 4 | True
C.	1 | False 2 | True 3 | True 4 | True
D.	1 | False 2 | True 3 | True 4 | True

Question 3:
What is the expected output of the following code?

```
data = {}
data['2'] = [1, 2]
data['1'] = [3, 4]

for i in data.keys():
    print(data[i][1], end=' ')
```

A. 1 3
B. 4 2
C. 3 1
D. 2 4

Question 4:
The ABC company needs a way to find the count of particular letters in their publications to ensure that there is a good balance. It seems that there have been complaints about overuse of the letter e .You need to create a function to meet the requirements.

```
# Function accepts a list of words from a file,
# and a letter to search for.
# Returns count of the words containing that letter.

def count_letter(letter, word_list):
    count = 0
    for ???:
        if ???:
            count += 1
    return count

word_list = []
# word_list is populated from a file. Code not shown.

letter = input('Which letter would you like to count?')
letter_count = count_letter(letter, word_list)
print('There are', letter_count, 'words with the letter', letter)
```

What would you insert instead of ??? and ???

A.	1 | word in word_list 2 | word in letter
B.	1 | word_list in word 2 | letter in word
C.	1 | word_list in word 2 | word in letter
D.	1 | word in word_list 2 | letter is word
E.	1 | word in word_list 2 | letter in word
F.	1 | word is word_list 2 | letter in word

Question 5:
Consider the following code.

```
for i in range(5, 0, ???):
    print(i, i, i, i, i)
```

What would you insert instead ??? so that the program prints the following pattern to the monitor?

```
5 5 5 5 5
4 4 4 4 4
3 3 3 3 3
2 2 2 2 2
1 1 1 1 1
```

A. 0
B. None
C. -1
D. 1

Question 6:
What is the expected result of the following code?

```
try:
    raise Exception
except:
    print("c")
except BaseException:
    print("a")
except Exception:
    print("b")
```

A. b
B. 1
C. a
D. The code will cause a syntax error.

Question 7:
Which of the following is an example of a Python built-in concrete exception?

A. IndexError
B. BaseException
C. ArithmeticError
D. ImportError

Question 8:
What is the expected output of the following code?

```
data = ['Peter', 'Paul', 'Mary']
print(data[int(-1 / 2)])
```

A. Peter
B. The code is erroneous.
C. Mary
D. Paul
E. None of the above.

Question 9:
You are writing an application that uses the sqrt function. The program must reference the function using the name squareRoot. You need to import the function.
Which code segment should you use?

A. from math.sqrt as squareRoot
B. import math.sqrt as squareRoot
C. from math import sqrt as squareRoot
D. import sqrt from math as squareRoot

Question 10:
Which of the following is the correct order of operator precedence?

A.	Parentheses Exponents Unary positive, negative, not Addition and Subtraction Multiplication and Division And
B.	Exponents Unary positive, negative, not Parentheses Multiplication and Division Addition and Subtraction And
C.	Exponents Unary positive, negative, not Multiplication and Division Addition and Subtraction And Parentheses
D.	Parentheses Exponents Unary positive, negative, not Multiplication and Division Addition and Subtraction And

Question 11:
The following code reads two numbers. Which of the following is the correct input for the code?

```
x, y = eval(input('Enter two numbers: '))
print(x)
print(y)
```

A. 3, 4

B. <pre>3 4</pre>
C. 3 4
D. None of the above.

Question 12:
Which of the following statements are correct? Choose two.

A. True and False evaluates to True
B. 7 + False evaluates to False
C. True + 1 evaluates to 2
D. True or false evaluates to True
E. type(‘ ‘) returns <class ‘bool’ >

Question 13:
You work for a company that distributes media for all ages. You are writing a function that assigns a rating based on a user's age. The function must meet the following requirements:
Anyone 18 years old or older receives a rating of A
Anyone 13 or older, but younger than 18 receives a rating of T
Anyone 12 years old or younger receives a rating of C
If the age is unknown, the rating is set to C

```
1  def get_rating(age):
2      rating = ''
3      if              # Line-3
4      elif            # Line-4
5      elif            # Line-5
6      else            # Line-6
7      return rating
```

Which of the following should you insert on Line-3 to Line-6?

A.	1 | : rating = 'A' # Line-3 2 | age < 13: rating = 'C' # Line-4 3 | age < 18: rating = 'T' # Line-5 4 | age == None: rating = 'C' # Line-6
B.	1 | : age < 18: rating ='"T' # Line-3 2 | age < 13: rating = 'C' # Line-4 3 | age == None: rating ='C" # Line-5 4 | : rating = 'A" # Line-6
C.	1 | age == None: rating = 'C' # Line-3 2 | age < 18: rating = 'C' # Line-4 3 | age < 13: rating = 'T' # Line-5 4 | : rating = 'A' # Line-6
D.	1 | : rating = 'A' # Line-3 2 | age < 18: rating = 'T' # Line-4 3 | age < 13: rating = 'C' # Line-5 4 | age == None: rating = 'C' # Line-6
E.	1 | age == None: rating = 'C' # Line-3 2 | age < 13: rating = 'C' # Line-4 3 | age < 18: rating = 'T' # Line-5 4 | age == None: rating = 'C' # Line-6
F.	1 | age < 13: rating = 'C' # Line-3 2 | age == None: rating = 'C' # Line-4 3 | age < 18: rating = 'T' # Line-5 4 | age == None: rating = 'C' # Line-6

Question 14:
The ABC company is building a basketball court for its employees to improve company morale. You are creating a Python program that employees can use to keep track of their average score. The program must allow users to enter their name and current scores. The program will output the user name and the user's average score. The output must meet the following requirements:
The user name must be left-aligned. If the user name has fewer than 20 characters, additional space must be added to the right. The average score must have three places to the left of the decimal point and one place to the right of the decimal (xxx.x)
What would you insert instead ??? and ???

```
name = input('What is your name?')
sum = 0
score = 0
count = 0
while score != -1:
    score = int(input('Enter your scores: (-1 to end)'))
    if score == -1:
        break
    sum += score
    count += 1
average = sum / count
print('???, your average score is: ???' % (name, average))
```

A.	1 | %-20% 2 | %4.1f
B.	1 | %-20f 2 | %4.1s
C.	1 | %-20s 2 | %1.4f
D.	1 | %-20d 2 | %4.1f
E.	1 | %-20i 2 | %4.1
F.	1 | %-20f 2 | %1.4s

Question 15:
What is the expected output of the following code?

```
data1 = (1, 2)
data2 = (3, 4)
[print(sum(x)) for x in [data1 + data2]]
```

A. Nothing gets printed.
B. 4
C. 10
D. The code is erroneous

Question 16:
What is the expected output of the following code?

```
x = 28
y = 8
print(x / y)
print(x // y)
print(x % y)
```

A.	1 | 3.5 2 | 3.5 3 | 2
B.	1 | 3.0 2 | 3 3 | 2
C.	1 | 3.5 2 | 3 3 | 4
D.	1 | 3 2 | 3.5 3 | 4

Question 17:
Consider the following code.

```
x = 42
y = 7
data = "I'm gonna make him an offer he can't refuse."
```

Which of the following expressions will evaluate to 19?

A. 19 if None else x / y
B. data.find('an') if data else None
C. 7 if len(data) > 19 else 6
D. data.rfind('an') if data else None

Question 18:
What does the readline() method return, when the end of the file is reached?

A. -1
B. EOF
C. An empty string.

D. An exception

Question 19:
You are creating a function that manipulates a number. The function has the following requirements: A float is passed into the function. The function must take the absolute value of the float. Any decimal points after the integer must be removed.
Which maths functions should you use? Each correct answer is part of the solution. Choose two.

A. math.frexp(x)
B. math.fmod(x)
C. math.floor(x)
D. math.ceil(x)
E. math.fabs(x)

Question 20:
The following statement ...

```
assert x != 0
```

A. will stop the program if x is not equal to 0
B. will stop the program if x is equal to 0
C. has no effect.
D. is erroneous.

Question 21:
What is the expected output of the following code?

```
print('x', 'y', 'z', sep='sep')
```

A. xsepysepzsep
B. xyz
C. x y z
D. xsepysepz

Question 22:

You want to write more text into an already existing file. Which of the modes below do you have to use to only open the file for writing?

A. ab+
B. at
C. at+
D. ab

Question 23:
Which of the following code snippets will print all prime numbers between 2 and 100 to the monitor?

A.

```
num = 2
while num <= 100:
    is_prime = True
    for i in range(2, num):
        is_prime = False
        break
    if is_prime == True:
        print(num)
```

B.

```
num = 2
is_prime = True
while num <= 100:
    for i in range(2, num):
        if num % i == 0:
            is_prime = False
            break
    if is_prime == True:
        print(num)
```

C.

```
num = 2
while num <= 100:
    is_prime = True
    for i in range(2, num):
        if num % i == 0:
            is_prime = False
            break
    if is_prime == False:
        print(num)
    num += 1
```

D.

```
num = 2
while num <= 100:
    is_prime = True
    for i in range(2, num):
        if num % i == 0:
            is_prime = False
            break
    if is_prime == True:
        print(num)
    num += 1
```

Question 24:
Consider the following code.

```
data = eval(input('Input: '))
print('Output:', data)
```

Which of the inputs below would produce the specified output?

A.	1 | Input: Hello Python 2 | Output: Hello Python
B.	1 | Input: [x**2 for x in range(1, 4)] 2 | Output: [1, 4, 9]
C.	Both are correct.
D.	None of the above.

Question 25:
What happens if you run the following code?

```
import time

time.sleep(30)
print("Wake up!")
```

A. The string Wake up! will be displayed thirty times on the screen for 30 seconds.
B. The string Wake up! will be displayed on the screen after 30 seconds.
C. The string Wake up! will be displayed on the screen for 30 seconds.
D. None of the above.

Question 26:
What is the expected output of the following code?

```
list = ['Peter', 'Paul', 'Mary']

def list(data):
    del data[1]
    data[1] = 'Jane'
    return data

print(list(list))
```

A. [‘Peter’ , ‘Jane’ , ‘ Mary’]
B. [‘Peter’ , ‘Jane’]
C. The code is erroneous.
D. [‘Paul’ , ‘Mary’, ‘Jane’]

Question 27:
Look at the code below:

```
import random

#
# Insert lines of code here.
#

print(a, b, c)
```

Which lines of code would you insert so that it is possible for the program to output the following result:

```
6 82 0
```

A.	1 | a = random.randrange(10, 100, 3) 2 | b = random.randint(0, 100) 3 | c = random.choice((0, 100, 3))
B.	1 | a = random.randint(0, 100) 2 | b = random.choice((0, 100, 3) 3 | c = random.randrange(10, 100, 3)
C.	1 | a = random.randint(0, 100) 2 | b = random.randrange(10, 100, 3) 3 | c = random.choice((0, 100, 3))
D.	1 | a = random.choice((0, 100, 3)) 2 | b = random.randrange(10, 100, 3) 3 | c = random.randint(0, 100)

Question 28:
Consider the following Python code:

```
distance = 1876.23
amount = +42E7
country = 'Italy'
```

What are the types of the variables distance, amount and country?

A. float, float, str
B. float, str, str
C. float, int, str
D. Double,

Question 29:
What is the expected output of the following code?

```
def func(x):
    if x % 2 == 0:
        return 1
    else:
        return 2

print(func(func(2)))
```

A. The code is erroneous
B. 0
C. 1
D. 2

Question 30:
What snippet would you insert in the line indicated below to print

```
The highest number is 10 and the lowest number is 1.
```

to the monitor?

```
data = [10, 2, 1, 7, 5, 6, 4, 3, 9, 8]
# insert your code here
print(
    ('The highest number is {} ' +
     'and the lowest number is {}.').format(high, low)
)
```

A.	None of the above.
B.	1 | def find_high_low(nums): 2 | nums.sort( ) 3 | return nums[len(nums)], nums[0] 4 | 5 | high, low = find_high_low(data)
C.	1 | def find_high_low(nums): 2 | nums.sort( ) 3 | return nums[0], nums[-1] 4 | 5 | high, low = find_high_low(data)
D.	1 | def find_high_low(nums): 2 | nums.sort( ) 3 | return nums[-1], nums[0] 4 | 5 | high, low = find_high_low(data)

Question 31:
You want to develop a program to calculate the area of a circle. Here is the formula:

```
A = pi * r ** 2
```

Which of the following code snippets do you need?

A.	None of the above.
B.	1 | import math 2 | def get_area(r): 3 | return math.pi * math.pow(r, 2)
C.	1 | import math 2 | def get_area(r): 3 | return math.pi * math.fmod(r, 2)
D.	1 | import math 2 | def get_area(r): 3 | return math.pi * math.fabs(r, 2)

Question 32:
If you want to build a string that reads:

```
Peter's sister's name's "Anna"
```

Which of the following literals would you use? Choose two.

A. 'Peter's sister's name's "Anna" '
B. "Peter's sister's name's "Anna" "
C. "Peter's sister's name's \ "Anna\ " "
D. 'Peter\ 's sister\ 's name\ 's \ "Anna\ " '

Question 33:
What is the expected output of the following code?

```
def func(x, y, z):
    return x + 4 * y + 5 * z

print(func(1, z=2, y=3))
```

A. 24
B. The code is erroneous.
C. 23
D. 80

Question 34:
Consider the following code.

```
languages = ['English', 'Spanish', 'German']
more_languages = ['English', 'Spanish', 'German']
extra_languages = more_languages
```

Which statement will print True to the monitor? Choose two.

A. print (languages is extra_languages)
B. print (languages == more_languages)
C. print (more_languages is extra_languages)
D. print (languages is more_languages)

Question 35:
You want to access the test.txt file and retrieve each line in it.

```
file = open('test.txt')
# insert code here
file.close()
```

Which option will you use? (Select all that apply.)

A.	print (file.readlines( ))
B.	print (file.lines( ))
C.	print (readlines(file))
D.	1 | for f in file: 2 | print(f)
E.	print (file.read( ))
F.	print (file.readlines(:)
G.	print (read.file(test.txt))

Question 36:
Consider the following code.

```
def function(x=0):
    return x
```

With sentence describes the function the best? A function defined like function() ...

A. may be called without any argument, or with just one.
B. must be called with exactly one argument.
C. must be called without arguments.
D. may be called with any number of arguments (including zero).

Question 37:
What is the expected output of the following code?

```
x = (1, 4, 7, 9, 10, 11)
y = {2: 'A', 4: 'B', 6: 'C', 8: 'D', 10: 'E', 12: 'F'}
res = 1

for z in x:
    if z in y:
        continue
    else:
        res += z

print(res)
```

A. 29
B. 28
C. 30
D. 31

Question 38:
What is the output of the following code?

```
import math

try:
    print(math.pow(2))
except TypeError:
    print('A')
else:
    print('B')
```

A. B
B. The program will cause a ValueError exception.
C. A
D. None of the above

Question 39:
What is the expected output of the following code?

```
data = (1, 2, 3, 4)
data = data[-2:-1]
data = data[-1]
print(data)
```

A. 2
B. 4
C. 3
D. 1

Question 40:
You are writing code that generates a random integer with a minimum value of 5 and a maximum value of 11
Which functions should you use? Each correct answer presents a complete solution. Choose two.

A. random.randrange(5, 11, 1)
B. random.randint(5, 11)
C. random.randrange(5, 12, 1)
D. random.randint(5, 12)

Question 41:
What is the expected output of the following code?

```
def func(n):
    s = '*'
    for i in range(n):
        s += s
    yield s

for x in func(2):
    print(x, end='')
```

A. *
B. ***
C. **
D. ****

Question 42:
What is the expected output of the following code?

```
print('Mike' > 'Mikey')
```

A. 1
B. True
C. False
D. 0

Question 43:
You execute the following command in the terminal.

```
python index.py Hello
```

You want the command to print out Hello
What has to be inside of index.py?

A.	1 | from sys import args 2 | print (args[1])
B.	1 | from sys import args 2 | print (args[0])
C.	1 | from sys import args 2 | print (args[0])
D.	1 | from sys import argv 2 | print (args[1])

Question 44:
What is the expected output of the following code?

```
data = {1: 0, 2: 1, 3: 2, 0: 1}
x = 0

for _ in range(len(data)):
    x = data[x]

print(x)
```

A. 1
B. The code is erroneous.
C. 0
D. 2

Question 45:
What is the expected output of the following code?

```
data = ['abc', 'def', 'abcde', 'efg']
print(max(data))
```

A. abcde
B. abc
C. def
D. The code is erroneous.
E. efg
F. None of the above

Question 46:
The following code:

```
print(float("1, 3"))
```

A. raises a ValueError exception.
B. prints 13
C. prints 1, 3
D. prints 1.3

Question 47:
Q636 - Basics
The compiled Python bytecode is stored in files having names ending with:

A. pyb
B. pyc
C. py
D. pc

Question 48:
Which expression evaluates to 7?

A. 9 % 3 + 7
B. 9 - 3 * 7
C. 9 / 3 * 7
D. 9 / / 3 - 7

Question 49:
Consider the following code.

```
start = input('How old were you at the time of joining?')
now = input('How old are you today?')
```

Which of the following codes will print the right output?

A.	1 | print(2 | ‘Congrats on ‘ 3 | + (int (now) - int (start)) 4 | + ‘ years of service !’ 5 |)
B.	1 | print(2 | ‘Congrats on ‘ 3 | + (int (now - start) 4 | + ‘ years of service !’ 5 |)
C.	1 | print(2 | ‘Congratulations on ‘ 3 | + str(int (now) - int (start)) 4 | + ‘ years of service !’ 5 |)
D.	1 | print(2 | ‘Congrats on ‘ 3 | + str(now - start) 4 | + ‘ years of service !’ 5 |)

Question 50:

You are designing a decision structure to convert a student's numeric grade to a letter grade. The program must assign a letter grade as specified as followed:

```
 90 through 100 -> A
 80 through  89 -> B
 70 through  79 -> C
 65 through  69 -> D
  0 through  64 -> F
```

For example, if the user enters a 90 the output should be Your letter grade is A
Likewise, if a user enters an 89 the output should be Your letter grade is B

```
# Letter Grade Converter
grade = int(input('Enter a numeric grade:'))
# Line-3
    letter_grade = 'A'
# Line-5
    letter_grade = 'B'
# Line-7
    letter_grade = 'C'
# Line-9
    letter_grade = 'D'
else:
    letter_grade = 'F'
print('Your letter grade is:', letter_grade)
```

Which of the following should you insert on Line-3, Line-5, Line-7 and Line-9?

A.	1 | if grade <= 90: # Line - 3 2 | elif grade <= 80: # Line - 5 3 | elif grade <= 70: # Line - 7 4 | elif grade <= 65: # Line - 9
B.	1 | if grade < 90: # Line - 3 2 | elif grade < 80: # Line - 5 3 | elif grade < 70: # Line - 7 4 | elif grade < 65: # Line - 9
C.	1 | if grade > 90: # Line - 3 2 | elif grade > 80: # Line - 5 3 | elif grade > 70: # Line - 7 4 | elif grade > 65: # Line - 9
D.	1 | if grade >= 90: # Line - 3 2 | elif grade >= 80: # Line - 5 3 | elif grade >= 70: # Line - 7 4 | elif grade >= 65: # Line - 9

Question 51:

What is the expected output of the following code?

```
def func(a, b):
    return a ** a

print(func(2))
```

A. None
B. 2
C. The code is erroneous.
D. 4

Question 52:
What is the expected output of the following code?

```
def func(x=2, y=3):
    return x * y

print(func(y=2, 3))
```

A. The code is erroneous.
B. 2
C. 4
D. 6

Question 53:
What is the expected output of the following code?

```
res = str(bool(1) + float(12) / float(2))
print(res)
```

A. 7.0
B. The code is erroneous.
C. 6.0
D. 6
E. 7

Question 54:
What is the expected output of the following code?

```
def func():
    print(x + 1, end=' ')

x = 1
func()
print(x)
```

A. 1 1
B. 1 2
C. 2 2
D. 2 1

Question 55:
What is the expected output of the following code if the user enters 4 and 7?

```
x = input()
y = int(input())
print(x * y)
```

A. 28
B. 4444444
C. 47
D. 7777

Question 56:
What is the expected output of the following code?

```
try:
    raise Exception
except:
    print('c')
except BaseException:
    print('a')
except Exception:
    print('b')
```

A. b
B. a
C. The code is erroneous.
D. c

Question 57:
The top-most Python exception is named:

A. TopException
B. PythonException
C. Exception
D. BaseException